AI Ethics For Lung Segmentation Model

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Thesis Title: A Hybrid Convolutional Neural Network-Vision Transformer Architecture with Parallel Feature Processing and Cross-Modal Attention for Lung Segmentation

Slide deck of proposal phase: (in experiements steps)

https://docs.google.com/presentation/d/1RsPmMdacFRiqHwna6HgGuuabOqbthgkj5YxTsj8C9D8/edit?usp=sharing

Instructions

This canvas is a tool to help you think systematically about the ethical dimensions and societal impact of your AI research project. Reflect on the principles discussed in the lecture (Fairness, Accountability, Transparency, etc.) and answer the questions in each section as they apply to your work. Be specific and thoughtful in your responses.

1. Stakeholders & Societal Impact

Who is affected by your project? Think beyond just the end-user.

Direct Users (Who will directly interact with your system?)	**Indirect Stakeholders (Who might be affected by the outcomes, even if they don't use it? e.g., society, community groups, competitors.)**
• Radiologists and medical imaging specialists who would use the segmentation tool for diagnosis • Healthcare institutions seeking automated lung segmentation systems • Researchers in medical image analysis and computer vision	• Patients whose chest X-rays would be analyzed by the system • Healthcare systems that could become more efficient with automated screening • Medical device manufacturers who might license the technology • Healthcare insurers and policy makers affected by diagnostic accuracy changes • Rural and underserved communities who might benefit from improved automated diagnosis

Intended Benefits (What is the primary positive impact you hope to achieve?)	**Potential Harms & Negative Consequences (What are the potential negative impacts, misuse scenarios, or unintended side effects? Think about the worst-case scenario.)**
• Improved accuracy and consistency in lung boundary detection for respiratory disease diagnosis • Reduced workload for radiologists, enabling faster diagnosis and treatment • Better early detection of respiratory diseases like tuberculosis, pneumonia, and lung cancer • Enhanced healthcare access in regions with limited radiological expertise	• Misdiagnosis if the system fails on edge cases or unseen pathologies • Over-reliance on automated systems, potentially reducing human expertise • Worsening of healthcare inequities if the system works better for certain demographics • False sense of security leading to missed critical diagnoses • Job displacement concerns for medical imaging technicians

2. Data & Bias

The foundation of your model. "Bias in, Bias out."

Data Sources & Collection (Where does your data come from? How was it collected and labeled?)	**Data Privacy & Security (Does your data contain personal or sensitive information? How will you protect it (e.g., anonymization, access controls)?)**
• Montgomery County dataset (138 chest X-rays from tuberculosis screening program) • Shenzhen Hospital dataset (662 frontal chest X-rays from Chinese medical institution) • Quatar-19 covid dataset (collected through a set chest xray images with annotation groudtruth from universities of Quatar • Manual annotation using Label Studio for ground truth creation	• Using publicly available, de-identified datasets that comply with medical data sharing protocols • No personal patient information included in the datasets • Secure storage and processing following university data protection policies • Anonymized data processing throughout the research pipeline

• Images collected from different populations (US and Chinese) with varying pathological conditions	

Potential Biases (What kinds of bias might exist in your dataset? Consider Representation Bias, Historical Bias, Measurement Bias, etc.)	**Bias Mitigation Strategy (What steps will you take to identify and reduce the impact of these biases? e.g., data augmentation, re-sampling, using fairness-aware algorithms.)**
• **Geographic bias**: Data from only a certain part of the world may not generalize globally • **Equipment bias**: Different X-ray machines and imaging protocols between datasets • **Demographic bias**: Limited age ranges and ethnic diversity in the datasets • **Pathological bias**: Focus on tuberculosis/covid-19 cases may not represent full spectrum of lung diseases • **Temporal bias**: Historical data may not reflect current imaging standards	• Cross-dataset validation to test generalization across different populations. Test on never seen data during training model • Data augmentation techniques to increase variability (rotation, contrast adjustment) • Documentation of dataset limitations and recommended use cases • Plans for future validation on more diverse datasets before clinical deployment

3. Model & Fairness

Inside the "black box." How does it make decisions?

Model Function & Logic (Briefly, what is the core task of your model? What is the input and what is the output?)	**Explainability (XAI) (On a scale of 1-5, how critical is it to explain this model's decisions? What methods might you use to explain it (e.g., SHAP, LIME, Decision Tree visualization)?)**
• Input: Chest X-ray images augmented to 512x512 pixels • Core task: Automated segmentation	• Medical applications require high explainability for clinical acceptance • Potential methods: Grad-CAM

of lung boundaries in chest radiographs • Architecture: Parallel CNN and Vision Transformer pathways with cross-modal attention fusion • Output: Binary masks indicating lung regions vs. background	visualization showing which image regions influenced the decision • Attention map visualization from the Vision Transformer component • Boundary confidence scoring to indicate segmentation certainty • Comparison with radiologist annotations to validate decision-making

Definition of "Fairness" for Your Project (What does a "fair" outcome look like in your specific context? e.g., equal loan approval rates, similar diagnostic accuracy across groups.)	**Fairness Assessment (How will you measure and test for fairness? What metrics will you use (e.g., Demographic Parity, Equal Opportunity)?)**
• Equal segmentation accuracy across different demographic groups (age, gender, ethnicity) • Consistent performance regardless of disease severity or pathological presentation • Similar accuracy across different imaging equipment and protocols • No systematic bias in lung boundary detection that could affect downstream diagnosis	• Training model with strict dataset protocal to avoid data leakage • Cross-dataset evaluation to assess generalization fairness • Statistical testing for significant performance differences between groups • Multiple evaluation metrics from pixel wise metrics, overlapping metrics to distance-based metrics for boundary of lungs to compare to SOTA models

4. Deployment & Governance

How will your project live in the real world?

Robustness & Safety (How could your model fail in the real world (e.g., unexpected data, adversarial attacks)? How will you test for and prevent this?)	**Accountability & Recourse (If the system makes a harmful mistake, who is responsible? Is there a process for users to appeal a decision or have it corrected?)**
• **Potential failures**: Poor performance on severely abnormal lung shapes, rare pathologies, or	• For research phase: Academic institution's ethics board oversight • For future deployment: Healthcare

low-quality images • **Adversarial concerns**: Susceptibility to image artifacts or intentional manipulation • **Testing strategies**: Extensive validation on edge cases, stress testing with corrupted images • **Safety measures**: Confidence scoring to flag uncertain segmentations for human review • **Fail-safe mechanisms**: Clear indicators when the system confidence is below acceptable thresholds	institution protocols for AI-assisted diagnosis appeals • Documentation of all model decisions and confidence levels for audit trails • Human oversight requirements for all clinical applications

Regulatory Compliance (What laws, standards, or policies must your project comply with (e.g., data privacy laws, medical device regulations, university ethics policies)?)	**Long-Term Monitoring (How would you monitor the model's performance and ethical impact after deployment to ensure it doesn't degrade or develop new biases over time?)**
• University of Technology ethics policies and research guidelines • International standards for medical device development (if commercialized) • Data protection laws in countries where datasets originated • Publication ethics guidelines for medical AI research • Future FDA or equivalent regulatory approval processes for clinical use	• Performance tracking across different patient populations over time • Regular evaluation against evolving clinical standards and practices • Monitoring for dataset drift as imaging technologies advance • Feedback collection from clinical users to identify bias emergence • Periodic re-validation on new, more diverse datasets as they become available • Version control and update protocols to maintain ethical standards